

# **Norway**

**SLEIPNER**

**Volve F**

**F-7**

**F-7**

**Plan: F7 Rev0.13 FinalCoords CC 040707**

## **Standard Survey Report**

**11 april, 2018**

# Statoil

## Survey Report

|                  |                                  |                                     |                                                     |
|------------------|----------------------------------|-------------------------------------|-----------------------------------------------------|
| <b>Company:</b>  | Norway                           | <b>Local Co-ordinate Reference:</b> | Site Volve F                                        |
| <b>Project:</b>  | SLEIPNER                         | <b>TVD Reference:</b>               | Rotary Table @ 54.90m (Rotary Table (As Installed)) |
| <b>Site:</b>     | Volve F                          | <b>MD Reference:</b>                | Rotary Table @ 54.90m (Rotary Table (As Installed)) |
| <b>Well:</b>     | F-7                              | <b>North Reference:</b>             | Grid                                                |
| <b>Wellbore:</b> | F-7                              | <b>Survey Calculation Method:</b>   | Minimum Curvature                                   |
| <b>Design:</b>   | F7 Rev0.13 FinalCoords CC 040707 | <b>Database:</b>                    | Production EDM P246N                                |

|                    |                               |                      |                             |
|--------------------|-------------------------------|----------------------|-----------------------------|
| <b>Project</b>     | SLEIPNER, Norway              |                      |                             |
| <b>Map System:</b> | Universal Transverse Mercator | <b>System Datum:</b> | Mean Sea Level              |
| <b>Geo Datum:</b>  | European 1950 - Mean          |                      | Using Well Reference Point  |
| <b>Map Zone:</b>   | Zone 31N (0 E to 6 E)         |                      | Using geodetic scale factor |

|                              |               |                     |                |                                    |
|------------------------------|---------------|---------------------|----------------|------------------------------------|
| <b>Site</b>                  | Volve F, 15/9 |                     |                |                                    |
| <b>Site Position:</b>        |               | <b>Northing:</b>    | 6,478,563.53 m | <b>Latitude:</b> 58° 26' 29.8068 N |
| <b>From:</b> Map             |               | <b>Easting:</b>     | 435,050.02 m   | <b>Longitude:</b> 1° 53' 14.9285 E |
| <b>Position Uncertainty:</b> | 0.00 m        | <b>Slot Radius:</b> | 13.200 in      | <b>Grid Convergence:</b> -0.95 °   |

|                             |              |         |                        |                |
|-----------------------------|--------------|---------|------------------------|----------------|
| <b>Well</b>                 | F-7          |         |                        |                |
| <b>Well Position</b>        | <b>+N/-S</b> | 1.96 m  | <b>Northing:</b>       | 6,478,565.48 m |
|                             | <b>+E/-W</b> | -1.11 m | <b>Easting:</b>        | 435,048.91 m   |
| <b>Position Uncertainty</b> | 0.00 m       |         | <b>Wellhead Depth:</b> | 91.00 m        |
|                             |              |         | <b>Water Depth:</b>    | 91.00 m        |

|                  |                   |                    |                        |                      |                            |
|------------------|-------------------|--------------------|------------------------|----------------------|----------------------------|
| <b>Wellbore</b>  | F-7               |                    |                        |                      |                            |
| <b>Magnetics</b> | <b>Model Name</b> | <b>Sample Date</b> | <b>Declination (°)</b> | <b>Dip Angle (°)</b> | <b>Field Strength (nT)</b> |
|                  | NETICREFERENCE    | 08.11.2005         | -2.54                  | 71.62                | 50,283                     |

|                          |                                  |                  |                      |                      |  |
|--------------------------|----------------------------------|------------------|----------------------|----------------------|--|
| <b>Design</b>            | F7 Rev0.13 FinalCoords CC 040707 |                  |                      |                      |  |
| <b>Audit Notes:</b>      |                                  |                  |                      |                      |  |
| <b>Version:</b>          | <b>Phase:</b>                    | PLAN             | <b>Tie On Depth:</b> | 145.90               |  |
| <b>Vertical Section:</b> | <b>Depth From (TVD) (m)</b>      | <b>+N/-S (m)</b> | <b>+E/-W (m)</b>     | <b>Direction (°)</b> |  |
|                          | 145.90                           | 1.96             | -1.11                | 255.36               |  |

|                            |               |                                        |                         |                    |  |
|----------------------------|---------------|----------------------------------------|-------------------------|--------------------|--|
| <b>Survey Tool Program</b> | <b>Date</b>   | 31.05.2010                             |                         |                    |  |
| <b>From (m)</b>            | <b>To (m)</b> | <b>Survey (Wellbore)</b>               | <b>Tool Name</b>        | <b>Description</b> |  |
| 145.90                     | 233.90        | F7 Rev0.13 FinalCoords CC 040707 (F-7) | Wellbore Surveyor, stat | Gyro Tool from GD  |  |
| 233.90                     | 1,083.50      | F7 Rev0.13 FinalCoords CC 040707 (F-7) | Wellbore Surveyor, stat | Gyro Tool from GD  |  |

|                           |                        |                    |                           |                  |                  |                             |                            |                           |                          |  |
|---------------------------|------------------------|--------------------|---------------------------|------------------|------------------|-----------------------------|----------------------------|---------------------------|--------------------------|--|
| <b>Planned Survey</b>     |                        |                    |                           |                  |                  |                             |                            |                           |                          |  |
| <b>Measured Depth (m)</b> | <b>Inclination (°)</b> | <b>Azimuth (°)</b> | <b>Vertical Depth (m)</b> | <b>+N/-S (m)</b> | <b>+E/-W (m)</b> | <b>Vertical Section (m)</b> | <b>Dogleg Rate (°/30m)</b> | <b>Build Rate (°/30m)</b> | <b>Turn Rate (°/30m)</b> |  |
| 145.90                    | 0.00                   | 0.00               | 145.90                    | 1.96             | -1.11            | 0.00                        | 0.000                      | 0.000                     | 0.000                    |  |
| 254.90                    | 0.00                   | 0.00               | 254.90                    | 1.96             | -1.11            | 0.00                        | 0.000                      | 0.000                     | 0.000                    |  |
| 300.90                    | 1.00                   | 275.00             | 300.90                    | 1.99             | -1.51            | 0.38                        | 0.652                      | 0.652                     | 0.000                    |  |
| 330.90                    | 0.00                   | 0.00               | 330.90                    | 2.01             | -1.77            | 0.62                        | 1.000                      | -1.000                    | 0.000                    |  |
| 532.90                    | 0.00                   | 0.00               | 532.90                    | 2.01             | -1.77            | 0.62                        | 0.000                      | 0.000                     | 0.000                    |  |
| 648.22                    | 9.61                   | 260.00             | 647.67                    | 0.34             | -11.28           | 10.24                       | 2.500                      | 2.500                     | 0.000                    |  |
| 817.39                    | 9.61                   | 260.00             | 814.47                    | -4.57            | -39.09           | 38.39                       | 0.000                      | 0.000                     | 0.000                    |  |
| 886.75                    | 9.29                   | 249.60             | 882.90                    | -7.53            | -50.04           | 49.73                       | 0.750                      | -0.138                    | -4.496                   |  |

# Statoil

## Survey Report

|                  |                                  |                                     |                                                     |
|------------------|----------------------------------|-------------------------------------|-----------------------------------------------------|
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| <b>Project:</b>  | SLEIPNER                         | <b>TVD Reference:</b>               | Rotary Table @ 54.90m (Rotary Table (As Installed)) |
| <b>Site:</b>     | Volve F                          | <b>MD Reference:</b>                | Rotary Table @ 54.90m (Rotary Table (As Installed)) |
| <b>Well:</b>     | F-7                              | <b>North Reference:</b>             | Grid                                                |
| <b>Wellbore:</b> | F-7                              | <b>Survey Calculation Method:</b>   | Minimum Curvature                                   |
| <b>Design:</b>   | F7 Rev0.13 FinalCoords CC 040707 | <b>Database:</b>                    | Production EDM P246N                                |

| Planned Survey     |                 |             |                    |           |           |                      |                     |                    |                   |
|--------------------|-----------------|-------------|--------------------|-----------|-----------|----------------------|---------------------|--------------------|-------------------|
| Measured Depth (m) | Inclination (°) | Azimuth (°) | Vertical Depth (m) | +N/-S (m) | +E/-W (m) | Vertical Section (m) | Dogleg Rate (°/30m) | Build Rate (°/30m) | Turn Rate (°/30m) |
| 1,083.50           | 9.29            | 249.60      | 1,077.07           | -18.60    | -79.81    | 81.33                | 0.000               | 0.000              | 0.000             |

| Design Targets                                                                                                            |               |              |          |           |           |              |             |                   |                  |
|---------------------------------------------------------------------------------------------------------------------------|---------------|--------------|----------|-----------|-----------|--------------|-------------|-------------------|------------------|
| Target Name<br>- hit/miss target<br>- Shape                                                                               | Dip Angle (°) | Dip Dir. (°) | TVD (m)  | +N/-S (m) | +E/-W (m) | Northing (m) | Easting (m) | Latitude          | Longitude        |
| Utsira Top F7 071106<br>- plan hits target center<br>- Point                                                              | 0.00          | 0.00         | 882.90   | -7.53     | -50.04    | 6,478,556.00 | 435,000.00  | 58° 26' 29.5369 N | 1° 53' 11.8524 E |
| Utsira Top F7 v2 071106<br>- plan misses target center by 51.98m at 886.75m MD (882.90 TVD, -7.53 N, -50.04 E)<br>- Point | 0.00          | 0.00         | 884.90   | 6.48      | -100.06   | 6,478,570.00 | 434,950.00  | 58° 26' 29.9627 N | 1° 53' 8.7555 E  |
| Utsira Base<br>- plan misses target center by 339.75m at 1083.50m MD (1077.07 TVD, -18.60 N, -79.81 E)<br>- Point         | 0.00          | 0.00         | 1,065.90 | 15.38     | -417.67   | 6,478,578.90 | 434,632.50  | 58° 26' 30.0798 N | 1° 52' 49.1724 E |

| Casing Points      |                    |         |                      |                    |  |
|--------------------|--------------------|---------|----------------------|--------------------|--|
| Measured Depth (m) | Vertical Depth (m) | Name    | Casing Diameter (in) | Hole Diameter (in) |  |
| 245.90             | 245.90             | 30"     | 30.000               | 36.000             |  |
| 919.90             | 915.61             | 13 3/8" | 13.375               | 17.500             |  |
| 1,083.33           | 1,076.90           | 9 5/8"  | 9.625                | 12.250             |  |

| Formations         |                    |             |           |         |                   |
|--------------------|--------------------|-------------|-----------|---------|-------------------|
| Measured Depth (m) | Vertical Depth (m) | Name        | Lithology | Dip (°) | Dip Direction (°) |
| 886.75             | 882.90             | Utsira Top  |           | 0.00    |                   |
| 1,073.20           | 1,066.90           | Utsira Base |           | 0.00    |                   |

Checked By: \_\_\_\_\_ Approved By: \_\_\_\_\_ Date: \_\_\_\_\_